



Cardiovascular Pharmacology

The 5-HT₁ receptors inhibiting the rat vasodepressor sensory CGRPergic outflow: Further involvement of 5-HT_{1F}, but not 5-HT_{1A} or 5-HT_{1D}, subtypesAbimael González-Hernández^a, Guadalupe Manrique-Maldonado^a, Jair Lozano-Cuenca^a, Enriqueta Muñoz-Islas^{a,1}, David Centurión^a, Antoinette Maassen VanDenBrink^b, Carlos M. Villalón^{a,*}^a Departamento de Farmacobiología, Cinvestav-Coapa, Cza. de los Tenorios 235, Col. Granjas-Coapa, Deleg. Tlalpan, 14330 México D.F., Mexico^b Division of Vascular Medicine and Pharmacology, Department of Internal Medicine, Erasmus MC, Rotterdam, P.O. Box 2040, 3000 CA Rotterdam, The Netherlands

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ABSTRACT

We have previously shown that 5-HT_{1B} receptors inhibit prejunctionally the rat vasodepressor CGRPergic sensory outflow. Since 5-HT₁ receptors comprise 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D} and 5-HT_{1F} functional subtypes, this study has further investigated the role of 5-HT_{1A}, 5-HT_{1D} and 5-HT_{1F} receptor subtypes in the inhibition of the above vasodepressor sensory outflow. Pithed rats were pretreated with i.v. continuous infusions of hexamethonium and methoxamine, followed by 5-HT₁ receptor agonists. Then electrical spinal stimulation (T₉–T₁₂) or i.v. bolus injections of exogenous α -CGRP produced frequency-dependent or dose-dependent vasodepressor responses. The electrically-induced vasodepressor responses remained unchanged during infusions of the 5-HT_{1A} receptor agonists 8-OH-DPAT and NN-DP-5-CT. In contrast, these responses were inhibited by the agonists sumatriptan (5-HT_{1A/1B/1D/1F}), indorenate (5-HT_{1A}), PNU-142633 (5-HT_{1D}) or LY344864 (5-HT_{1F}), which did not affect the vasodepressor responses to exogenous CGRP (implying a prejunctional sensory-inhibition). When analysing the effects of antagonists: (i) 310 μ g/kg (but not 100 μ g/kg) GR127935 (5-HT_{1A/1B/1D/1F}) abolished the inhibition to sumatriptan, indorenate, PNU-142633 or LY344864; (ii) 310 μ g/kg SB224289 (5-HT_{1B}) or BRL15572 (5-HT_{1D}) failed to block the inhibition to sumatriptan or PNU-142633, whereas SB224289 + BRL15572 partly blocked the inhibition to sumatriptan; and (iii) 10 μ g/kg WAY100635 (5-HT_{1A}) failed to block the inhibition to indorenate. These results suggest that 5-HT_{1F}, but not 5-HT_{1A} or 5-HT_{1D}, receptor subtypes inhibit the vasodepressor sensory CGRPergic outflow although, admittedly, no selective 5-HT_{1F} receptor agonist is available yet. The pharmacological profile of these receptors resembles that shown in rat dorsal root ganglia by molecular biology techniques.

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1. Introduction

The physiological role of calcitonin gene-related peptide (CGRP) in the cardiovascular system has been widely documented (see Brain and Grant, 2004; Deng and Li, 2005). In this respect: (i) resistance blood vessels are innervated by sympathetic (see Hoffman, 2001) and primary sensory CGRPergic (Taguchi et al., 1992) nerves, which modulate the vascular tone; and (ii) perivascular CGRPergic nerves are more abundant around the arteries than around the veins (Bell and McDermott, 1996). Indeed, electrical stimulation of primary sensory nerves originating from the spinal cord in pithed rats results in vasodepressor responses mediated by CGRP release (Taguchi et al., 1992).

Despite the importance of CGRP, few studies have investigated the receptors involved in the modulation of CGRPergic transmission in the systemic vasculature (e.g. Villalón et al., 2008). For example, sumatriptan (a 5-HT_{1A/1B/1D/1F} receptor agonist) inhibits prejunctionally the vasodepressor CGRPergic sensory outflow in pithed rats (Lozano-Cuenca et al., 2009), but no attempt was made to identify the receptors involved. This prejunctional inhibition is also produced by CP-93,129 and involves 5-HT_{1B} receptors (González-Hernández et al., 2010). Since 5-HT₁ receptors consist of 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D} and 5-HT_{1F} functional subtypes (see Villalón and Centurión, 2007), this study has further investigated the role of the 5-HT_{1A}, 5-HT_{1D} and 5-HT_{1F} subtypes in the above sensory-inhibition by using several 5-HT₁ receptor ligands (see Table 1) including: (a) the agonists sumatriptan (5-HT_{1A/1B/1D/1F}); indorenate, NN-DP-5-CT and 8-OH-DPAT (all 5-HT_{1A}); PNU-142633 (5-HT_{1D}); or LY344864 (5-HT_{1F}) in their capacity to inhibit the vasodepressor sensory CGRPergic outflow or the vasodepressor responses to i.v. bolus injections of exogenous α -CGRP; and (b) the antagonists GR127935 (5-HT_{1A/1B/1D/1F}), WAY100635 (5-HT_{1A}), SB224289 (5-HT_{1B}) and/or BRL15572 (5-HT_{1D}) in their capacity to block the inhibition induced by some of the above agonists.

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Table 1
Binding affinity constants (pK_i) of the various 5-HT₁ receptor agonists and antagonists used in the present study.

	5-HT _{1A}	5-HT _{1B}	5-HT _{1D}	5-HT _{1E}	5-HT _{1F}
Agonists					
Sumatriptan	6.4 ^a	(r)7.3 ^b	8.5 ^a	5.8 ^a	7.9 ^a
Indorenate	7.8 ^c	5.4 ^c	6.7 ^c	–	–
8-OH-DPAT	(p)8.7 ^d	(r)5.1 ^b	7.4 ^e	5.5 ^f	5.8 ^f
NN-DP-5-CT	8.7 ^g	5.8 ^g	6.6 ^g	–	–
PNU-142633	–	(g)4.8 ^h	(g)8.3 ^h	–	–
LY344864	6.3 ⁱ	6.3 ⁱ	6.2 ⁱ	–	8.2 ⁱ
Antagonists					
GR127935	7.2 ^j	(r)8.8 ^b	8.6 ^j	5.4 ^j	6.4 ^j
WAY100635	9.6 ^k	5.1 ^k	5.6 ^k	–	–
SB224289	5.5 ^l	8.0 ^l	6.2 ^l	<5.0 ^l	<5.0 ^l
BRL15572	7.7 ^j	6.1 ^j	7.9 ^j	5.2 ^j	6.0 ^j

All data are given as pK_i values at human receptors, except when stated otherwise: gorilla (g), porcine (p) and rodent (r) receptors.

^a Laysen et al., 1996.

^b Beer et al., 1998.

^c Hoyer D (personal communication).

^d Hoyer, 1989.

^e Pauwels et al., 1996.

^f Adham et al., 1993.

^g Hoyer et al., 1994.

^h Pregoner et al., 1999.

ⁱ Phebus et al., 1997.

^j Price et al., 1997.

^k Fletcher et al., 1993.

^l Hagan et al., 1997.

2. Materials and methods

2.1. Animals

All animal protocols in this investigation were approved by our Institutional Ethics Committee (CICUAL-Cinvestav), in accordance with the Guide for the Care and Use of Laboratory Animals in U.S.A. Male Wistar normotensive rats (300–350 g) were maintained at a 12/12-h light/dark cycle (light beginning at 07:00 h) and housed at constant temperature (22 ± 2 °C) and humidity (50%), with food and water *ad libitum*.

2.2. General methods

Experiments were carried out in a total of 273 rats. After quick anaesthesia with diethyl ether and cannulation of the trachea, the rats were pithed by inserting a stainless steel rod through the orbit and foramen magnum into the vertebral foramen (Gillespie et al., 1970). Then, the animals were artificially ventilated with room air using a model 7025 Ugo Basile pump (56 strokes/min; stroke volume: 20 ml/kg), as established by Kleinman and Radford (1964). After bilateral vagotomy, catheters were placed in: (i) the left and right femoral and jugular veins, for the continuous infusions of agonists (methoxamine followed by 5-HT₁ receptor agonists), and i.v. administration of 5-HT₁ receptor antagonists, respectively; and (ii) the left carotid artery, connected to a Grass pressure transducer (P23XL), for the recording of blood pressure. Heart rate was measured with a tachograph (7P4, Grass Instrument Co., Quincy, MA, U.S.A.), triggered from the blood pressure signal. Both parameters were recorded by a model 7 Grass polygraph (Grass Instrument Co., Quincy, MA, U.S.A.). At this point, the 273 rats were divided into two main sets (see Fig. 1), so that the effects produced by the infusions of methoxamine followed by sumatriptan, indorenate, NN-DP-5-CT, 8-OH-DPAT, PNU-142633 or LY344864 under different treatments could be investigated on the vasodepressor responses induced by: (i) electrical stimulation of the perivascular (vasodepressor) sensory outflow (set 1; $n=243$); or (ii) i.v. bolus injections of exogenous α -CGRP (set 2; $n=30$). The stimulus–response curves or dose–response curves elicited by electrical stimulation and

exogenous α -CGRP were completed in about 50 min under unaltered values of resting blood pressure. The electrical stimuli (0.56, 1, 1.8, 3.1 and 5.6 Hz), as well as the doses of α -CGRP (0.1, 0.18, 0.31, 0.56 and 1 μ g/kg), were given using a sequential schedule at 5–10 min intervals (see below), as previously reported (Lozano-Cuenca et al., 2009). The body temperature of each pithed rat was maintained at 37 °C by a lamp and monitored with a rectal thermometer.

2.3. Experimental protocols

After the animals ($n=273$) had been in a stable haemodynamic condition for at least 15 min, baseline values of diastolic blood pressure (a more accurate indicator of peripheral vascular resistance) and heart rate were determined.

2.3.1. Electrical stimulation of the perivascular (vasodepressor) sensory outflow

In the first set of rats ($n=243$; see Fig. 1), the pithing rod was replaced by an electrode enamelled except for 1.5 cm length 9 cm from the tip, so that the uncovered segment was situated at T₉–T₁₂ of the spinal cord, and an indifferent electrode was placed dorsally (Taguchi et al., 1992; Villalón et al., 2008). Before electrical stimulation, the animals received (i.v.): (i) a bolus injection of gallamine (25 mg/kg) to avoid the electrically-induced muscular twitching; (ii) 10 min later, a continuous infusion of hexamethonium (2 mg/kg.min) to block the electrically-induced vasopressor responses that are produced by stimulation of the preganglionic sympathetic vasopressor outflow (Villalón et al., 2008); and (iii) 10 min later, a continuous infusion of methoxamine (20 μ g/kg.min) to increase diastolic blood pressure. Then, this set of rats was divided into 3 groups ($n=81$, 50 and 112, respectively; see Fig. 1).

The first group ($n=81$) was divided into 8 subgroups that received i.v. continuous infusions of: (i) saline (0.02 ml/min; $n=5$); (ii) 1, 10 and 100 μ g/kg.min sumatriptan ($n=5$ each); (iii) 10, 31 and 100 μ g/kg.min indorenate ($n=5$ each); (iv) 1 and 3.1 μ g/kg.min NN-DP-5-CT ($n=5$ and 3, respectively); (v) 3.1 and 10 μ g/kg.min 8-OH-DPAT ($n=5$ and 3, respectively); (vi) 1, 10 and 100 μ g/kg.min PNU-142633 ($n=5$ each); (vii) 10 and 31 μ g/kg.min LY344864 ($n=5$ each); or (viii) nothing (control group; $n=5$). After 20 min, diastolic blood pressure and heart rate were determined again; then the perivascular sensory outflow was electrically stimulated during the above treatments to elicit vasodepressor responses by applying 10-s trains of monophasic, rectangular pulses (2 ms, 50 V), at the above frequencies. When diastolic blood pressure had returned to baseline levels, the next frequency was applied. This procedure was systematically performed until the stimulus–response curve had been completed.

The second group ($n=50$) received an i.v. continuous infusion of saline (0.02 ml/min) and after 10 min was divided into 8 subgroups comprising i.v. injections of: (i) saline (1 ml/kg; $n=4$); (ii) 2.5% and 5% dimethylsulphoxide (DMSO) (1 ml/kg; $n=4$ each); (iii) GR127935 (100 and 310 μ g/kg; $n=5$ each); (iv) WAY100635 (10 and 31 μ g/kg; $n=5$ each); (v) SB224289 (310 μ g/kg; $n=4$); (vi) BRL15572 (310 μ g/kg; $n=4$); (vii) SB224289 + BRL15572 (310 μ g/kg each; $n=5$); and (viii) WAY100635 + GR127935 (10 + 100 μ g/kg, respectively; $n=5$). After 10 min, a stimulus–response curve was elicited to analyse the effect of these antagonists *per se*.

The third group ($n=112$) was divided into 4 subgroups comprising i.v. continuous infusions of, respectively: (i) sumatriptan (10 μ g/kg.min; $n=48$); (ii) indorenate (31 μ g/kg.min; $n=20$); (iii) PNU-142633 (10 μ g/kg.min; $n=29$); and (iv) LY344864 (31 μ g/kg.min; $n=15$). After 10 min, each of these subgroups was subdivided as follows. The sumatriptan subgroup ($n=48$) was subdivided into 8 subgroups comprising i.v. injections of: (i) saline (1 ml/kg, control; $n=5$); (ii) 2.5% and 5% DMSO (1 ml/kg; $n=4$ each); (iii) GR127935 (100 and 310 μ g/kg; $n=5$ each); (iv) SB224289 (310 μ g/kg; $n=5$); (v) BRL15572 (310 μ g/kg; $n=5$); (vi) SB224289 + BRL15572 (310 μ g/kg each; $n=5$);

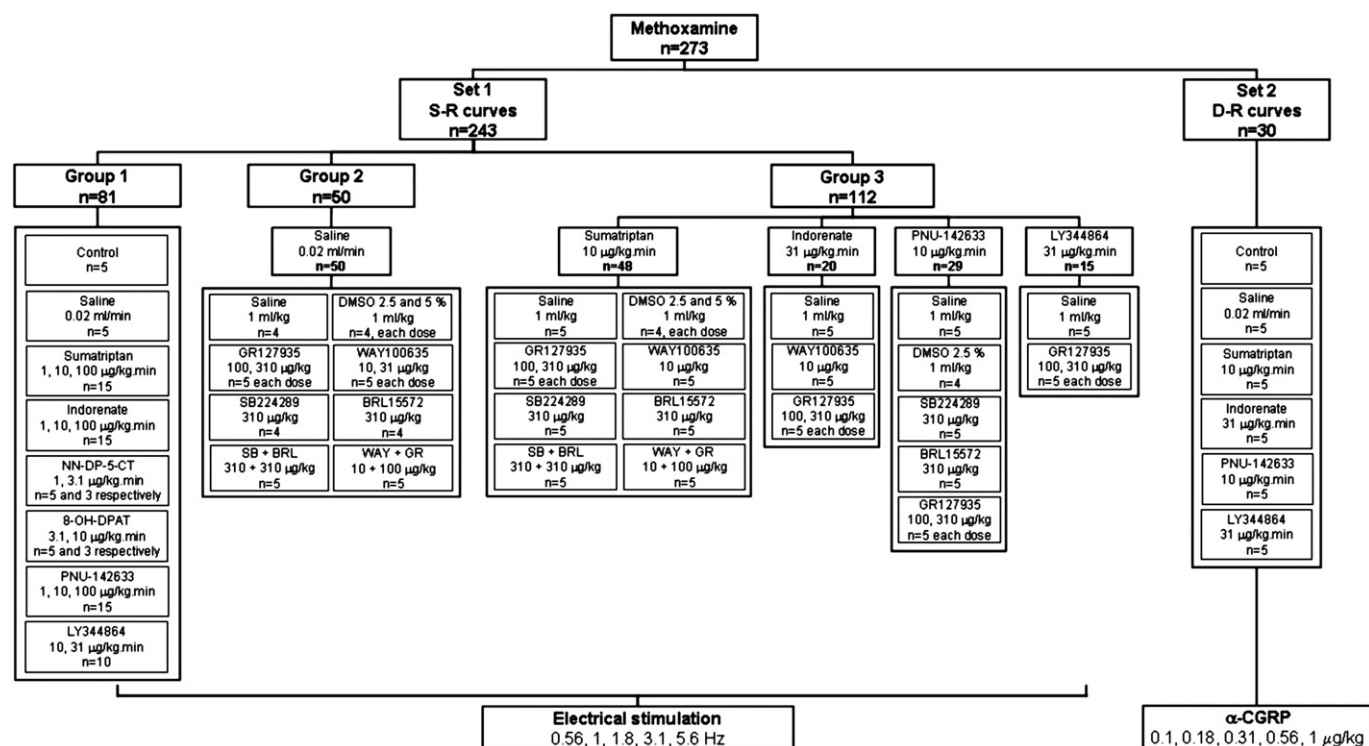


Fig. 1. Experimental protocols showing the number of animals used in the two main sets and the different groups and subgroups used in the present study. Methoxamine was infused during and until the end of the experiment. S–R, stimulus–response curve and D–R, dose–response curve to exogenous α -CGRP. WAY, WAY100635; SB, SB224289; and BRL, BRL15572.

(vii) WAY100635 (10 μ g/kg; $n = 5$); or (viii) WAY100635 + GR127935 (10 + 100 μ g/kg, respectively; $n = 5$). The indorenate subgroup ($n = 20$) was subdivided into 3 subgroups: (i) saline (1 ml/kg; $n = 5$); (ii) WAY100635 (10 μ g/kg; $n = 5$); and (iii) GR127935 (100 and 310 μ g/kg; $n = 5$ each). The PNU-142633 subgroup ($n = 29$) was subdivided into 5 subgroups comprising: (i) saline (1 ml/kg; $n = 5$); (ii) 2.5% DMSO (1 ml/kg; $n = 4$); (iii) SB224289 (310 μ g/kg; $n = 5$); (iv) BRL15572 (310 μ g/kg; $n = 5$); and (v) GR127935 (100 and 310 μ g/kg; $n = 5$ each). Finally, the LY344864 subgroup ($n = 15$) was subdivided into 2 subgroups receiving: (i) saline (1 ml/kg; $n = 5$); and (ii) GR127935 (100 and 310 μ g/kg; $n = 5$ each). In all cases, 10 min after antagonist treatment a stimulus–response curve was constructed as described above during the infusion of the 5-HT₁ receptor agonists. It should be pointed out that only one stimulus–response curve was carried out per animal since tachyphylaxis of the vasodepressor responses was observed when eliciting a second stimulus–response curve, as previously reported (Villalón et al., 2008; Lozano-Cuenca et al., 2009).

2.3.2. Administration of exogenous α -CGRP

The second set of rats ($n = 30$; see Fig. 1) was prepared as described above, but the pinning rod was left throughout the experiment and the administration of both gallamine and hexamethonium was omitted, as described earlier (Villalón et al., 2008; Lozano-Cuenca et al., 2009). This set received an i.v. continuous infusion of methoxamine (20 μ g/kg.min); after 10 min (when diastolic blood pressure was maintained at around 135 mm Hg), this set was divided into 6 groups ($n = 5$ each) that received (i.v.): (i) nothing (control group); (ii) saline (0.02 ml/min); (iii) sumatriptan (10 μ g/kg.min); (iv) indorenate (31 μ g/kg.min); (v) PNU-142633 (10 μ g/kg.min); or (vi) LY344864 (31 μ g/kg.min). Twenty minutes later, the values of diastolic blood pressure and heart rate were determined again and the vasodepressor responses elicited by i.v. bolus injections of exogenous α -CGRP (0.1, 0.18, 0.31, 0.56 and 1 μ g/kg) were examined in these groups during the infusions of agonists.

2.4. Other procedures applying to the experimental protocols (2.3.1 and/or 2.3.2)

Methoxamine and the 5-HT₁ receptor agonists were infused at a rate of 0.02 ml/min by a WPI model sp100i pump (World Precision Instruments Inc., Sarasota, FL, U.S.A.). Diastolic blood pressure was increased in all cases with a methoxamine infusion (20 μ g/kg.min). The interval between the different frequencies of stimulation or doses of α -CGRP depended on the duration of the resulting vasodepressor responses (5–10 min), as we waited until diastolic blood pressure had returned to baseline values.

2.5. Drugs

Apart from the anaesthetic (diethyl ether), the compounds used in this study (obtained from the sources indicated) were: gallamine triethiodide, hexamethonium chloride, rat α -CGRP, methoxamine hydrochloride, 2,3,6,7-tetrahydrospiro-1'-methyl-5-[2'-methyl-4'-(5-methyl-1,2,4-oxadiazol-3-yl)biphenyl-4-carbonyl] furo[2,3-f]indole-3-spiro-4'-piperidine hydrochloride (SB224289), 1-(3-chlorophenyl)-4-(3,3-diphenyl(2-(S,R)hydroxy propano) piperazine) hydrochloride (BRL15572), N-[2-[4-(2-methoxy-phenyl)-1-piperazinyl]ethyl]-N-(2-pyridinyl)cyclohexanecarboxamide trihydrochloride (WAY100635), N, N-dipropyl-5-carboxamidotryptamine maleate (NN-DP-5-CT) and 8-hydroxy-di-n-propylamino tetralin (8-OH-DPAT) (Sigma Chemical Co., St Louis, MO, U.S.A.); N-[(3R)-3-(dimethylamino)-2,3,4,9-tetrahydro-1H-carbazol-6-yl]-4-fluorobenzamide (LY344864), and (1S)-1-[2-[4-(aminocarbonyl)phenyl]-1-piperazinyl]ethyl]-3,4-dihydro-N-methyl-1H-2-benzopyran-6-carboxamide (PNU-142633) (Tocris Bioscience, Northpoint, U.K.); sumatriptan and N-[4-methoxy-3-(4-methyl-1-piperazinyl)phenyl]-2'-methyl-4'-(5-methyl-1,2,4-oxadiazol-3-yl)[1,1-biphenyl]-4-carboxamide hydrochloride monohydrate (GR127935) (gifts from GlaxoSmithKline, Stevenage, Hertfordshire, U.K.); 5-methoxytryptamine- β -methylcarboxylate hydrochloride (indorenate) (gift from Professor Enrique Hong, Cinvestav-IPN, Mexico City, Mexico).

All compounds were dissolved in saline, except: (i) SB224289 or BRL15572 (2.5% DMSO); and (ii) the combination of SB224289 + BRL15572 (5% DMSO). The resulting solutions were finally gauged with distilled water to avoid precipitation. These vehicles had no effect on baseline diastolic blood pressure or heart rate (see below). Fresh solutions were prepared for each experiment. The doses of antagonists refer to their respective salts, whereas those of the agonists refer to their free base.

2.6. Data presentation and statistical evaluation

All data in the text, tables and figures are presented as mean $\pm$ SEM. The peak changes in diastolic blood pressure by electrical stimulation or exogenous α -CGRP were expressed as percent change from baseline at the steady state effect of methoxamine. The difference in the absolute values of diastolic blood pressure and heart rate within one subgroup of animals before and during the continuous infusions of methoxamine (20 μ g/kg.min), and/or the 5-HT₁ receptor agonists (at the doses mentioned above), were evaluated with paired Student's *t*-test. Moreover, a one-way analysis of variance was used to compare the absolute values of diastolic blood pressure and heart rate obtained: (i) before and during the continuous infusions of methoxamine (20 μ g/kg.min), and/or the 5-HT₁ receptor agonists in the different subgroups; or (ii) during the continuous infusions of methoxamine (20 μ g/kg.min), and/or the 5-HT₁ receptor agonists before, immediately after and 10 min after administration of vehicles (saline or DMSO) or the 5-HT₁ receptor antagonists given alone or in combination. Finally, the vasodepressor responses to electrical stimulation or exogenous α -CGRP in the different subgroups of animals were compared with a two-way analysis of variance. The one- and two-way analysis of variance were followed, if applicable, by the Student–Newman–Keuls' post hoc test. Statistical significance was accepted at $P < 0.05$.

3. Results

3.1. Systemic haemodynamic effects produced by the different treatments

Baseline values of diastolic blood pressure and heart rate were 58 ± 4 mm Hg and 244 ± 6 beats per min, respectively ($n = 273$); these values remained unchanged after gallamine or hexamethonium (not shown). Twenty minutes after starting the infusion of methoxamine, diastolic blood pressure and heart rate were significantly increased in all cases. Table 2 shows this fact in the animals receiving infusions of vehicle or the 5-HT₁ receptor agonists for the analysis of the electrically-induced vasodepressor responses. Similar effects of methoxamine during infusions of these compounds were obtained for the analysis of the vasodepressor responses to exogenous α -CGRP (not shown). The absolute values of diastolic blood pressure and heart rate in the different subgroups before, immediately after and 10 min after i.v. saline, DMSO (2.5% or 5%) or antagonists during methoxamine infusion were not significantly different ($P > 0.05$) (Table 3). Except when constructing stimulus- or dose-response curves (see below), the increases in diastolic blood pressure and heart rate were sustained throughout the experiments (Fig. 2).

3.2. Vasodepressor responses to electrical stimulation or exogenous α -CGRP

Fig. 2 shows some experimental tracings illustrating that, during the continuous infusion of methoxamine, the onset of the responses to electrical stimulation (Fig. 2A) or i.v. bolus injections of exogenous α -CGRP (Fig. 2B) was immediate and resulted in frequency- or dose-dependent decreases in diastolic blood pressure. In all cases, the former vasodepressor responses were due to selective vasodepressor stimulation, since only negligible and inconsistent effects in heart rate were observed (Fig. 2A), as shown by Lozano-Cuenca et al. (2009). Interestingly, the sumatriptan infusion (10 μ g/kg.min) attenuated the

Table 2

Baseline values of diastolic blood pressure and heart rate before and 20 min after administration of an i.v. continuous infusion of methoxamine and/or vehicle (saline) or the several 5-HT₁ receptor agonists used in the present study.

Treatment	Doses (μ g/kg.min)	n	Diastolic blood pressure (mm Hg)		Heart rate (beats per min)	
			Before	After	Before	After
Methoxamine (control)	20	5	48 $\pm$ 6	141 $\pm$ 8 ^b	249 $\pm$ 6	285 $\pm$ 6 ^b
Saline ^a	0.02 ml/min	5	53 $\pm$ 3	133 $\pm$ 7 ^b	245 $\pm$ 6	293 $\pm$ 7 ^b
Sumatriptan ^a	1	5	48 $\pm$ 7	143 $\pm$ 9 ^b	249 $\pm$ 4	280 $\pm$ 5 ^b
	10	5	48 $\pm$ 3	139 $\pm$ 7 ^b	267 $\pm$ 2	296 $\pm$ 3 ^b
	100	5	47 $\pm$ 5	128 $\pm$ 4 ^b	241 $\pm$ 3	279 $\pm$ 9 ^b
Indorenate ^a	10	5	55 $\pm$ 7	153 $\pm$ 6 ^b	239 $\pm$ 2	283 $\pm$ 7 ^b
	31	5	52 $\pm$ 3	149 $\pm$ 10 ^b	257 $\pm$ 5	296 $\pm$ 3 ^b
	100	5	58 $\pm$ 5	138 $\pm$ 6 ^b	261 $\pm$ 4	289 $\pm$ 7 ^b
8-OH-DPAT ^a	3.1	5	55 $\pm$ 7	123 $\pm$ 4 ^b	259 $\pm$ 6	275 $\pm$ 5 ^b
	10	3	58 $\pm$ 3	149 $\pm$ 11 ^b	257 $\pm$ 5	296 $\pm$ 4 ^b
NN-DP-5-CT ^a	1	5	55 $\pm$ 5	128 $\pm$ 7 ^b	249 $\pm$ 7	279 $\pm$ 6 ^b
	3.1	3	59 $\pm$ 7	133 $\pm$ 9 ^b	255 $\pm$ 7	287 $\pm$ 7 ^b
PNU-142633 ^a	1	5	49 $\pm$ 7	133 $\pm$ 13 ^b	243 $\pm$ 6	305 $\pm$ 4 ^b
	10	5	48 $\pm$ 3	139 $\pm$ 5 ^b	252 $\pm$ 2	296 $\pm$ 5 ^b
	100	5	51 $\pm$ 7	143 $\pm$ 8 ^b	239 $\pm$ 9	270 $\pm$ 7 ^b
LY344864 ^a	10	5	48 $\pm$ 3	129 $\pm$ 9 ^b	262 $\pm$ 4	296 $\pm$ 6 ^b
	31	5	53 $\pm$ 5	138 $\pm$ 11 ^b	251 $\pm$ 5	289 $\pm$ 7 ^b

^a During a continuous infusions of methoxamine (20 μ g/kg.min).

^b $P < 0.05$, after vs. before from the corresponding baseline value (paired *t*-test). All values are expressed as mean $\pm$ SEM.

vasodepressor responses to electrical stimulation (Fig. 2A), but not those to i.v. α -CGRP (Fig. 2D); similar results were obtained during infusions of indorenate, PNU-142633 or LY344864 (not shown). In contrast, the NN-DP-5-CT infusion (3.1 μ g/kg.min) had no effect on the responses to electrical stimulation (Fig. 2E).

In all cases (Fig. 2), the vasodepressor responses to electrical stimulation or i.v. α -CGRP: (i) appeared about 10 s after starting each stimulus/dose and reached a maximum 1 min after the stimulus had ended; and (ii) returned to baseline levels within 5–10 min after each stimulus/dose, as reported by Villalón et al. (2008). This response profile was also produced during the infusion of the other agonists. It is noteworthy that during infusion of a greater dose of methoxamine (i.e. 30 μ g/kg.min), which produced a higher and sustained increase in diastolic blood pressure, electrical stimulation produced similar ($P > 0.05$) frequency-dependent vasodepressor responses (Villalón et al., 2008).

3.3. Vasodepressor responses to electrical stimulation or i.v. bolus injections of exogenous α -CGRP during continuous infusions of several 5-HT₁ receptor agonists

As shown in Fig. 3, during methoxamine infusion (control; 20 μ g/kg.min), the electrically-induced vasodepressor responses were: (i) unaltered during the infusion of vehicle (Fig. 3A), NN-DP-5-CT (Fig. 3D) or 8-OH-DPAT (Fig. 3E); and (ii) dose-dependently inhibited during infusions of sumatriptan (Fig. 3B), indorenate (Fig. 3C), PNU-142633 (Fig. 3F) or LY344864 (Fig. 3G). This inhibition was significant at 1.8, 3.1 and 5.6 Hz in all cases (see Fig. 3B, C, F and G).

In addition, i.v. injections of α -CGRP elicited dose-dependent vasodepressor responses during methoxamine infusion (control; Fig. 4A). These responses, unlike those by electrical stimulation (see above), remained unchanged during the infusions of vehicle (saline, 0.02 ml/min; Fig. 4A), sumatriptan (10 μ g/kg.min; Fig. 4B), indorenate (31 μ g/kg.min; Fig. 4C), PNU-142633 (10 μ g/kg.min; Fig. 4D) or LY344864 (31 μ g/kg.min; Fig. 4E). Since these doses of agonists inhibited the electrically-induced vasodepressor responses without significantly affecting those by exogenous α -CGRP, the above doses were chosen for further pharmacological analysis. It should be noted that 8-OH-DPAT and NN-DP-5-CT were not investigated here because they failed to inhibit the electrically-induced vasodepressor responses (see above).

Table 3

Values of diastolic blood pressure and heart rate during the infusion of methoxamine (20 $\mu\text{g}/\text{kg}\cdot\text{min}$): (i) before; (ii) immediately after; and (iii) 10 min after i.v. administration of saline, 2.5% DMSO, 5% DMSO or the antagonists used in the present study.

Treatment	Doses ($\mu\text{g}/\text{kg}$)	n	Diastolic blood pressure (mm Hg)			Heart rate (beats per min)		
			Before	Immediately after	10 min after	Before	Immediately after	10 min after
Saline	1 ^a	4	137 $\pm$ 8	143 $\pm$ 7	138 $\pm$ 9	302 $\pm$ 9	307 $\pm$ 7	298 $\pm$ 8
2.5% DMSO	1 ^a	4	143 $\pm$ 8	165 $\pm$ 3	140 $\pm$ 5	265 $\pm$ 9	249 $\pm$ 7	268 $\pm$ 15
5% DMSO	1 ^a	4	154 $\pm$ 15	165 $\pm$ 14	162 $\pm$ 12	284 $\pm$ 12	267 $\pm$ 13	284 $\pm$ 14
GR127935	100	4	132 $\pm$ 6	136 $\pm$ 6	143 $\pm$ 9	296 $\pm$ 9	283 $\pm$ 11	304 $\pm$ 12
	310	4	135 $\pm$ 8	139 $\pm$ 7	136 $\pm$ 7	292 $\pm$ 5	275 $\pm$ 8	285 $\pm$ 13
WAY100635	10	5	139 $\pm$ 5	135 $\pm$ 7	138 $\pm$ 8	267 $\pm$ 5	265 $\pm$ 7	267 $\pm$ 7
	31	5	127 $\pm$ 8	131 $\pm$ 11	130 $\pm$ 5	279 $\pm$ 7	265 $\pm$ 5	277 $\pm$ 5
WAY100635 + GR127935	10, 100	5	143 $\pm$ 3	140 $\pm$ 9	138 $\pm$ 9	302 $\pm$ 6	301 $\pm$ 5	302 $\pm$ 7
SB224289	310	4	140 $\pm$ 9	150 $\pm$ 6	149 $\pm$ 8	302 $\pm$ 9	312 $\pm$ 8	303 $\pm$ 12
BRL15572	310	4	135 $\pm$ 6	125 $\pm$ 9	135 $\pm$ 7	266 $\pm$ 13	268 $\pm$ 10	274 $\pm$ 11
SB224289 + BRL15572	310, 310	5	133 $\pm$ 9	138 $\pm$ 13	125 $\pm$ 8	312 $\pm$ 14	311 $\pm$ 15	314 $\pm$ 12

All values are expressed as mean $\pm$ SEM.

^a Saline and DMSO were given at a dose of 1 ml/kg.

3.4. Effect of vehicles or antagonists per se on the electrically-induced vasodepressor responses

Fig. 5 illustrates that during continuous infusion of methoxamine (20 $\mu\text{g}/\text{kg}\cdot\text{min}$), the vasodepressor responses to electrical stimulation in control animals did not significantly differ from those elicited in

animals pretreated (i.v.) with: (i) saline (1 ml/kg; Fig. 5A) or DMSO (2.5% and 5%, 1 ml/kg; Fig. 5B); or (ii) the antagonists GR127935 (5-HT_{1B/1D}, 100 and 310 $\mu\text{g}/\text{kg}$; Fig. 5C), WAY100635 (5-HT_{1A}, 10 $\mu\text{g}/\text{kg}$; Fig. 5D), SB224289 (5-HT_{1B}, 310 $\mu\text{g}/\text{kg}$; Fig. 5E), BRL15572 (5-HT_{1D}, 310 $\mu\text{g}/\text{kg}$; Fig. 5F), SB224289 + BRL15572 (310 $\mu\text{g}/\text{kg}$ each; Fig. 5G) or WAY100635 + GR127935 (10 and 100 $\mu\text{g}/\text{kg}$ respectively; Fig. 5H).

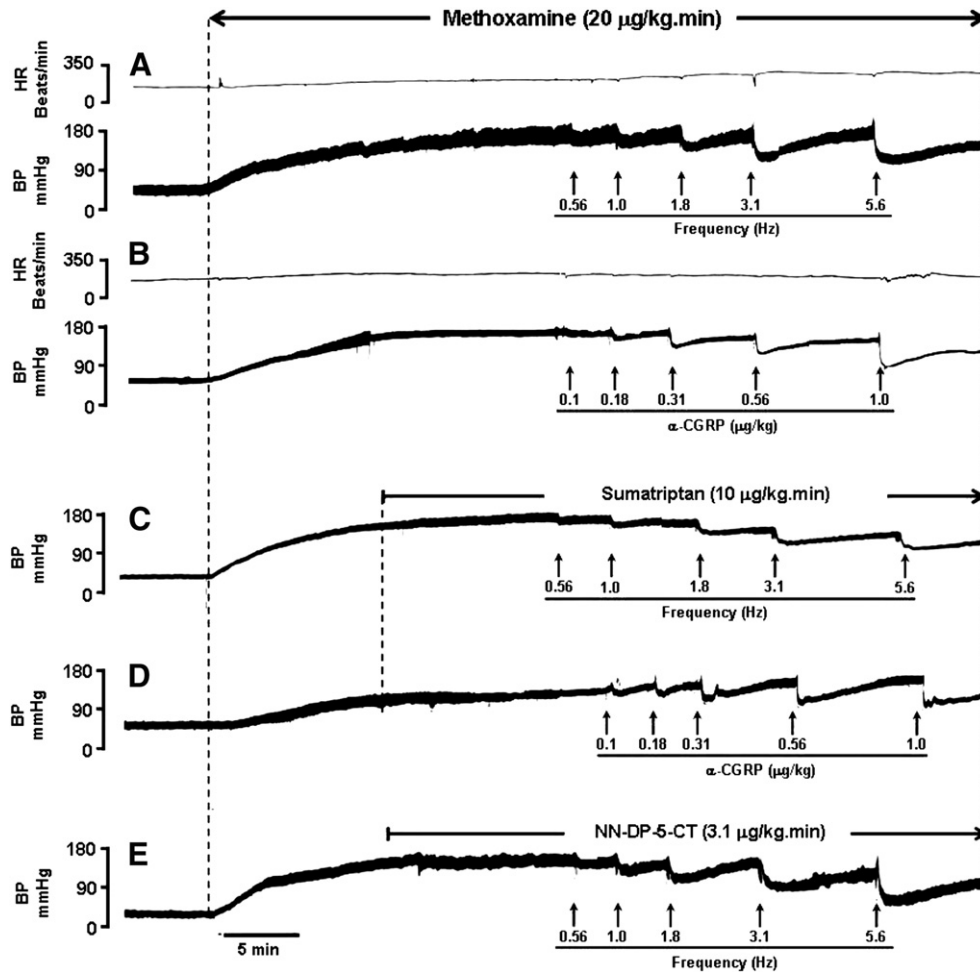


Fig. 2. Original tracings illustrating the vasodepressor responses to electrical stimulation of the perivascular sensory outflow or i.v. bolus injections of exogenous α -CGRP. Panels A and B show the effect of, respectively, electrical stimulation and α -CGRP on diastolic blood pressure under methoxamine infusion (control); in both cases the vasodepressor responses were not accompanied by changes in heart rate. Panels C and D depict the effect of sumatriptan infusion on the vasodepressor responses produced by electrical stimulation or α -CGRP. Panel E illustrates that NN-DP-5-CT had no effect on the vasodepressor responses to electrical stimulation. Note that only sumatriptan inhibited the vasodepressor responses to electrical stimulation (Panel C); similar results were obtained during infusions of indorenate, PNU-142633 or LY344864 (not shown).

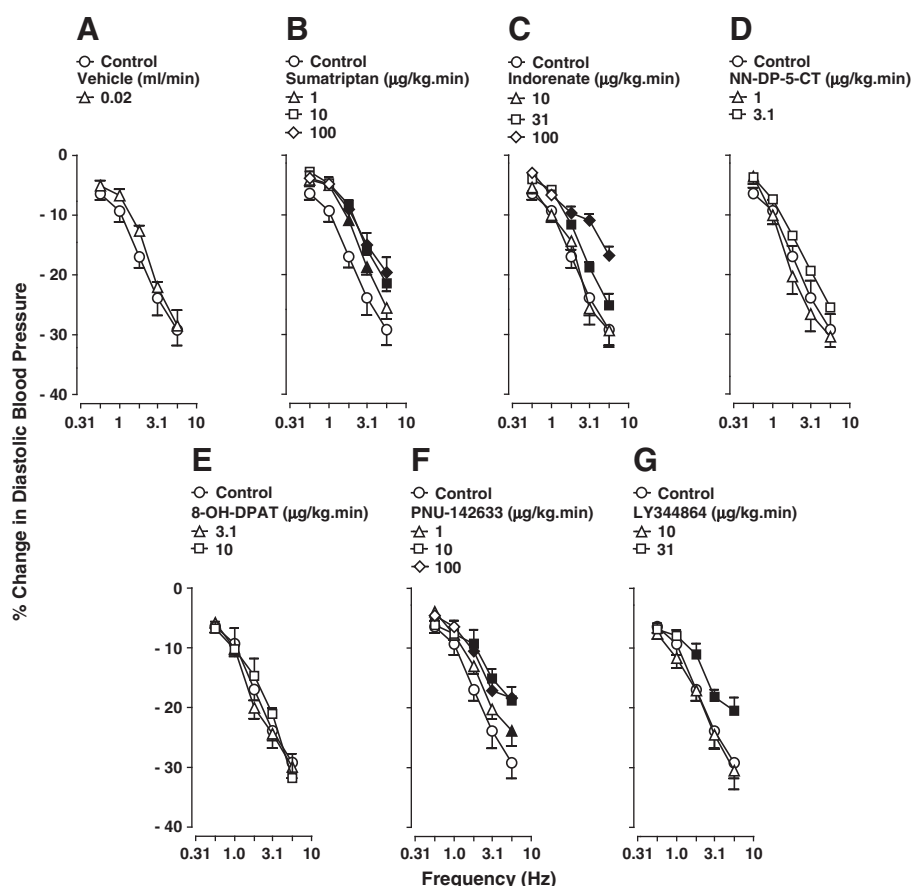


Fig. 3. Vasodepressor responses to electrical stimulation during i.v. continuous infusions of: (A) vehicle (saline; 0.02 ml/min; $n = 5$); (B) sumatriptan (1, 10 and 100 $\mu\text{g/kg.min}$; $n = 5$ each dose); (C) indoreinate (10, 31 and 100 $\mu\text{g/kg.min}$; $n = 5$ each dose); (D) NN-DP-5-CT (1 and 3.1 $\mu\text{g/kg.min}$; $n = 5$ and 3 respectively); (E) 8-OH-DPAT (3.1 and 10 $\mu\text{g/kg.min}$; $n = 5$ and 3 respectively); (F) PNU-142633 (1, 10 and 100 $\mu\text{g/kg.min}$; $n = 5$ each dose); or (G) LY344864 (10 and 31 $\mu\text{g/kg.min}$; $n = 5$ each dose). For the sake of clarity, empty symbols depict either control (methoxamine; 20 $\mu\text{g/kg.min}$) responses ($\circ$) or non-significant responses ($\triangle$, $\square$, $\diamond$) vs. control. Solid symbols ($\blacktriangle$, $\blacksquare$, $\blacklozenge$) represent significantly different responses ($P < 0.05$) vs. control; moreover, the inhibition produced by the highest doses of sumatriptan, indoreinate, PNU-142633 and LY344864 were more marked ($P < 0.05$) than those produced by their preceding doses, particularly at 5.6 Hz.

These findings indicate that the above compounds (except 31 $\mu\text{g/kg}$ WAY100635; Fig. 5D) at the doses used were devoid of any effects *per se* on the electrically-induced vasodepressor responses.

3.5. Effect of i.v. pretreatment with vehicles or antagonists on sumatriptan-, indoreinate-, PNU-142633- or LY344864-induced vascular sensory-inhibition

In animals pretreated with 1 ml/kg vehicle (saline, 2.5% DMSO or 5% DMSO), the inhibition of vasodepressor sensory outflow by infusions of

sumatriptan (10 $\mu\text{g/kg.min}$), indoreinate (31 $\mu\text{g/kg.min}$), PNU-142633 (10 $\mu\text{g/kg.min}$) or LY344864 (31 $\mu\text{g/kg.min}$) remained practically unaltered ($P > 0.05$) (see Figs. 6A and B, 7A, 8A and B and 9A).

Sumatriptan-induced inhibition was: (i) unaffected by SB224289 (310 $\mu\text{g/kg}$; Fig. 6D), BRL15572 (310 $\mu\text{g/kg}$; Fig. 6E) or WAY100635 (10 $\mu\text{g/kg}$; Fig. 6G); (ii) partially blocked by GR127935 (100 $\mu\text{g/kg}$; Fig. 6C), SB224289 + BRL15572 (310 $\mu\text{g/kg}$ each; Fig. 6F) or WAY100635 + GR127935 (10 and 100 $\mu\text{g/kg}$; Fig. 6H); and (iii) abolished by a supramaximal dose of GR127935 (310 $\mu\text{g/kg}$; Fig. 6C).

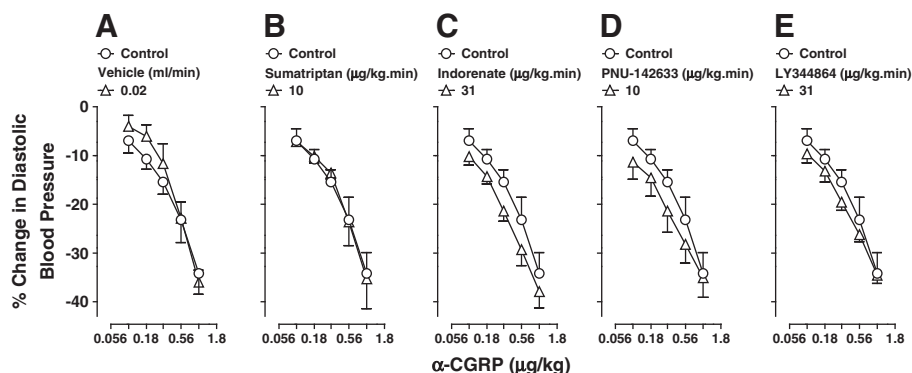


Fig. 4. Vasodepressor responses induced by i.v. bolus injections of exogenous α -CGRP elicited during i.v. continuous infusions of: (A) vehicle (saline; 0.02 ml/min); (B) sumatriptan (10 $\mu\text{g/kg.min}$); (C) indoreinate (31 $\mu\text{g/kg.min}$); (D) PNU-142633 (10 $\mu\text{g/kg.min}$); or (E) LY344864 (31 $\mu\text{g/kg.min}$) ($n = 5$ each dose).

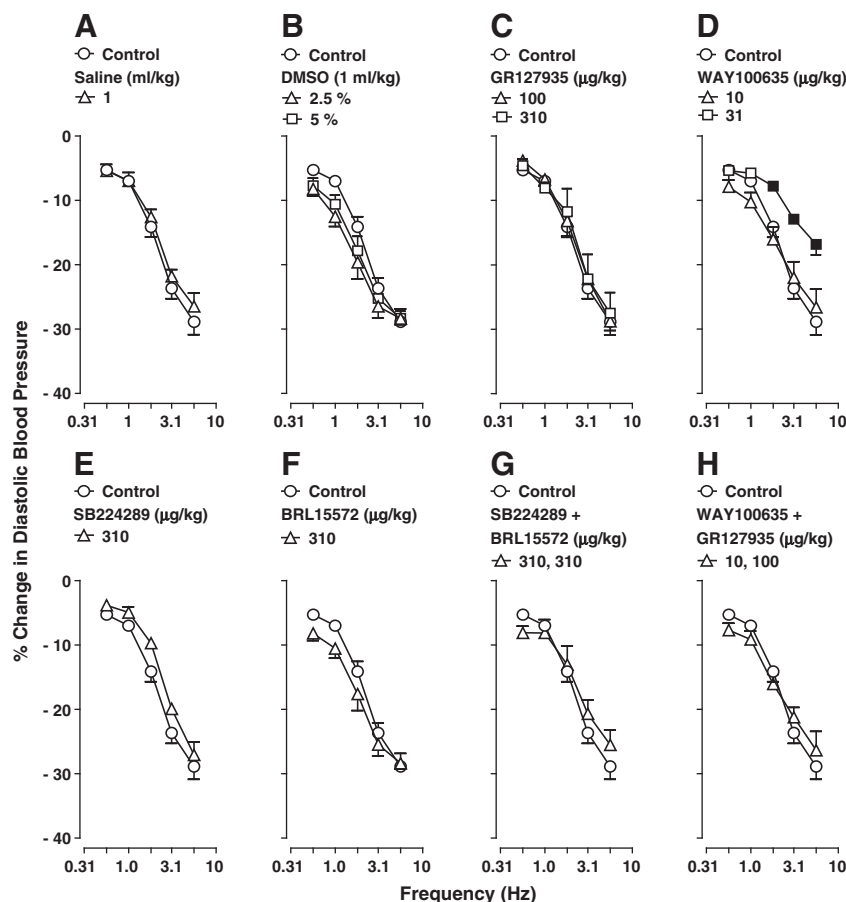


Fig. 5. Effect per se of i.v. bolus injections of: (A) saline (1 ml/kg; $n = 5$); (B) 2.5% DMSO or 5% DMSO (1 ml/kg; $n = 4$ each dose); (C) GR127935 (100 and 310 $\mu\text{g/kg}$; $n = 5$ each dose); (D) WAY100635 (10 and 31 $\mu\text{g/kg}$; $n = 5$ each dose); (E) SB224289 (310 $\mu\text{g/kg}$; $n = 4$); (F) BRL15572 (310 $\mu\text{g/kg}$; $n = 4$); (G) SB224289 plus BRL15572 (310 $\mu\text{g/kg}$ each; $n = 5$); or (H) WAY100635 plus GR127935 (10 + 100 $\mu\text{g/kg}$, respectively; $n = 5$). For the sake of clarity, empty symbols depict either control (methoxamine) responses ($\circ$) or non-significant responses ($\triangle$) vs. control. Solid symbols ($\blacksquare$) represent significantly different responses ($P < 0.05$) vs. control.

Furthermore, indorenate-induced inhibition was: (i) unaltered after WAY100635 (10 $\mu\text{g/kg}$; Fig. 7B) or GR127935 (100 $\mu\text{g/kg}$; Fig. 7C); and (ii) completely blocked by a supramaximal dose of GR127935 (310 $\mu\text{g/kg}$; Fig. 7C). PNU-142633-induced inhibition was: (i) unaffected after BRL15572 (310 $\mu\text{g/kg}$; Fig. 8C), SB224289 (310 $\mu\text{g/kg}$; Fig. 8D) or GR127935 (100 $\mu\text{g/kg}$; Fig. 8E); and (ii) abolished by a supramaximal dose of GR127935 (310 $\mu\text{g/kg}$; Fig. 8E). Finally, LY344864-induced inhibition was only reverted by 310 $\mu\text{g/kg}$ (but not by 100 $\mu\text{g/kg}$) GR127935 (Fig. 9B).

It is important to note that the above doses of antagonists (with the exception of 310 $\mu\text{g/kg}$ GR127935), were high enough to completely and *selectively* block their respective receptors mediating cardiovascular responses in pithed rats (Villalón et al., 1998; Sánchez-López et al., 2004; González-Hernández et al., 2010). Admittedly, our suggestion for supporting or excluding the role of the different 5-HT₁ receptor subtypes in this study is based on the assumption that species differences between the binding of the several ligands used do not play a major role (see Table 1).

4. Discussion

4.1. General

Regarding our experimental model, it is noteworthy that we did not measure sensory nerve activity directly, but the electrically-

induced neurotransmitter release (i.e. CGRP) was estimated indirectly by measurement of the evoked vasodepressor response. The responses to the various 5-HT₁ receptor agonists were considered sensory-inhibitory as they inhibited the vasodepressor responses to electrical stimulation, but not those to exogenous α -CGRP. Our results suggest that, apart from prejunctional 5-HT_{1B} receptors, the 5-HT₁ receptors inhibiting the vasodepressor sensory outflow resemble the pharmacological profile of the 5-HT_{1F}, but not of the 5-HT_{1A} and 5-HT_{1D} subtypes.

4.2. Systemic haemodynamic effects by the different treatments

As previously established in pithed rats to produce vasodepressor responses (Villalón et al., 2008), diastolic blood pressure was maintained at around 135 mm Hg by a methoxamine infusion. This sustained vasopressor response (Table 2) results from an increase in peripheral vascular resistance (see Hoffman, 2001) due to stimulation of vascular α_1 -adrenoceptors (Decker et al., 1984). Moreover, the increase in heart rate by methoxamine (Table 2), previously observed in similar studies (Villalón et al., 2008), could be due to activation of cardiac α_1 -adrenoceptors (Terzic et al., 1993).

The fact that sumatriptan, indorenate, 8-OH-DPAT, NN-DP-5-CT, PNU-142633 or LY344864 did not alter the vasopressor response to methoxamine infusion (Table 2) reflects that activation of peripheral 5-HT₁ receptors does not exert significant effects on blood pressure under our experimental conditions (Villalón and Centurión, 2007).

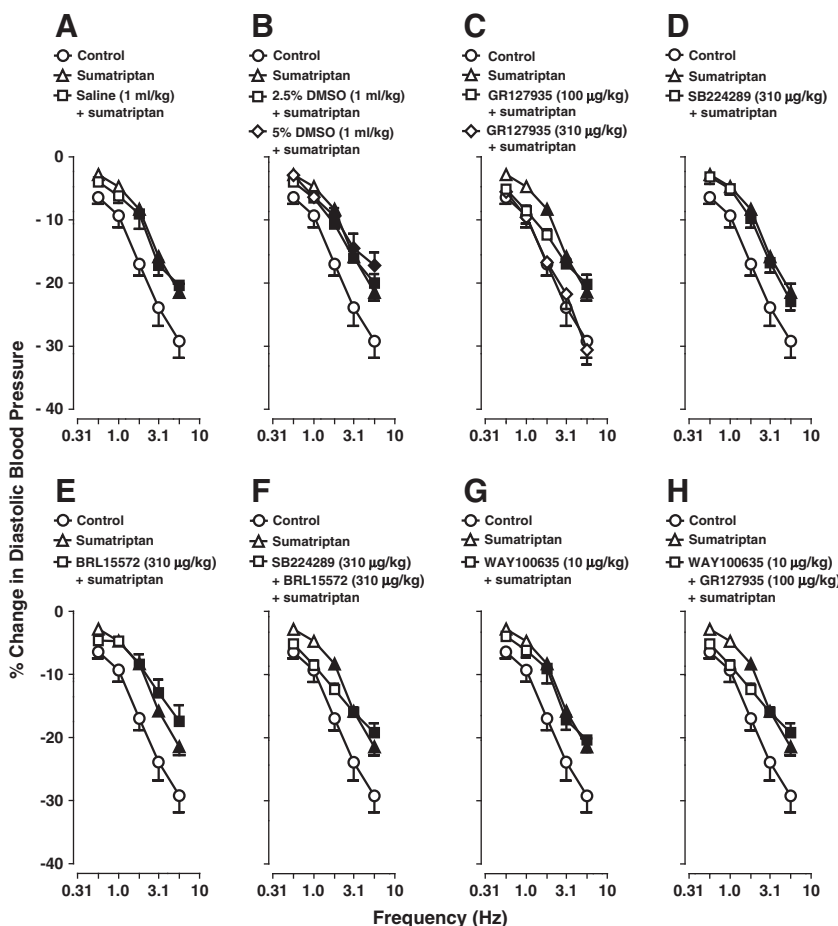


Fig. 6. Effect of i.v. bolus injections of: (A) saline (1 ml/kg; $n = 5$); (B) 2.5% DMSO or 5% DMSO (1 ml/kg; $n = 4$ each dose); (C) GR127935 (100 and 310 $\mu\text{g/kg}$; $n = 5$ each dose); (D) SB224289 (310 $\mu\text{g/kg}$; $n = 5$ each dose); (E) BRL15572 (310 $\mu\text{g/kg}$; $n = 5$ each dose); (F) SB224289 plus BRL15572 (310 $\mu\text{g/kg}$ each; $n = 5$ each dose); (G) WAY100635 (10 $\mu\text{g/kg}$; $n = 5$ each dose); or (H) WAY100635 plus GR127935 (10 + 100 $\mu\text{g/kg}$ respectively; $n = 5$), on sumatriptan-induced inhibition (10 $\mu\text{g/kg}\cdot\text{min}$) of the electrically-induced vasodepressor responses. The control group represents that of animals receiving an i.v. methoxamine infusion (20 $\mu\text{g/kg}\cdot\text{min}$), which is shown for comparison. For the sake of clarity, empty symbols depict either control (methoxamine) responses ($\circ$) or non-significant responses ($\triangle$, $\square$) vs. control. Solid symbols ($\blacktriangle$) represent significantly different responses ($P < 0.05$) vs. control.

4.3. The role of 5-HT₁ receptors in the sensory-inhibition to sumatriptan

Our results suggest that sumatriptan inhibited prejunctionally the vasodepressor sensory outflow because it inhibited the vasodepressor responses to electrical stimulation (Figs. 2C and 3B) without affecting those to α -CGRP (Fig. 4B). The role of 5-HT₁ receptors is established by the profile of blockade produced by GR127935, a 5-HT_{1A/1B/1D/1F} receptor antagonist (Table 1). This compound: (i) partly attenuated sumatriptan-induced inhibition (Fig. 6C) at 100 $\mu\text{g/kg}$, a dose that completely blocks prejunctional 5-HT_{1B} (pK_i : 8.8) and 5-HT_{1D} (pK_i : 8.6) receptors in pithed rats (Sánchez-López et al., 2004); and (ii) abolished sumatriptan-induced inhibition at the supramaximal dose of 310 $\mu\text{g/kg}$ (Fig. 6C), which may be blocking, considering its affinity profile (Table 1), prejunctional 5-HT_{1A} (pK_i : 7.2) and 5-HT_{1F} (pK_i : 6.4) receptors (in addition to 5-HT_{1B} and 5-HT_{1D} receptors; see above). The simplest interpretation of these findings may suggest (but does not prove) the role of 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D} and 5-HT_{1F} receptors but, admittedly, the situation is far more complex than hypothesised here. Many possibilities can be proposed, but some of the most relevant ones will be touched upon here. Accordingly, the combination of the antagonists SB224289 (5-HT_{1B}; 310 $\mu\text{g/kg}$, pK_i : 8.0) plus BRL15572 (5-HT_{1D}; 310 $\mu\text{g/kg}$, pK_i : 7.9) exhibited the same blocking profile (Fig. 6F) as 100 $\mu\text{g/kg}$ GR127935 (Fig. 6C), suggesting the role of 5-HT_{1B} and 5-HT_{1D} receptors. However, the fact that no blockade was produced when either SB224289 or BRL15572 was given alone (Fig. 6D and E) implies that the blockade of only one receptor is overshadowed by the non-blocked

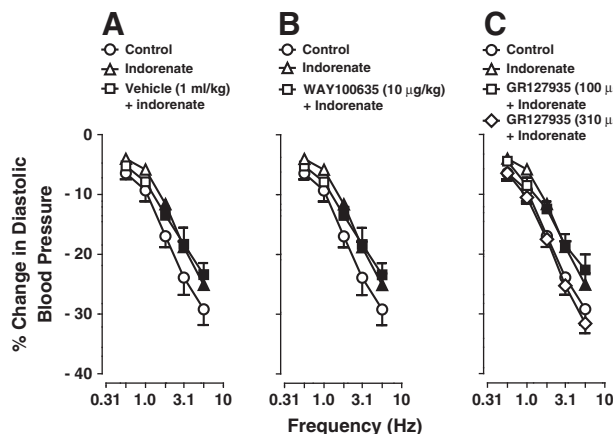


Fig. 7. Effect of i.v. bolus injections of: (A) vehicle (saline, 1 ml/kg); (B) WAY100635 (10 $\mu\text{g/kg}$); or (C) GR127935 (100 and 310 $\mu\text{g/kg}$) on the inhibition induced by indorenate (31 $\mu\text{g/kg}\cdot\text{min}$) ($n = 5$ each dose) of the electrically-induced vasodepressor responses. The control group represents that of animals receiving an i.v. continuous infusion of methoxamine (20 $\mu\text{g/kg}\cdot\text{min}$), which is shown for comparison. For the sake of clarity, empty symbols depict either control (methoxamine) responses ($\circ$) or non-significant responses ($\triangle$, $\square$) vs. control. Solid symbols ($\blacktriangle$) represent significantly different responses ($P < 0.05$) vs. control.

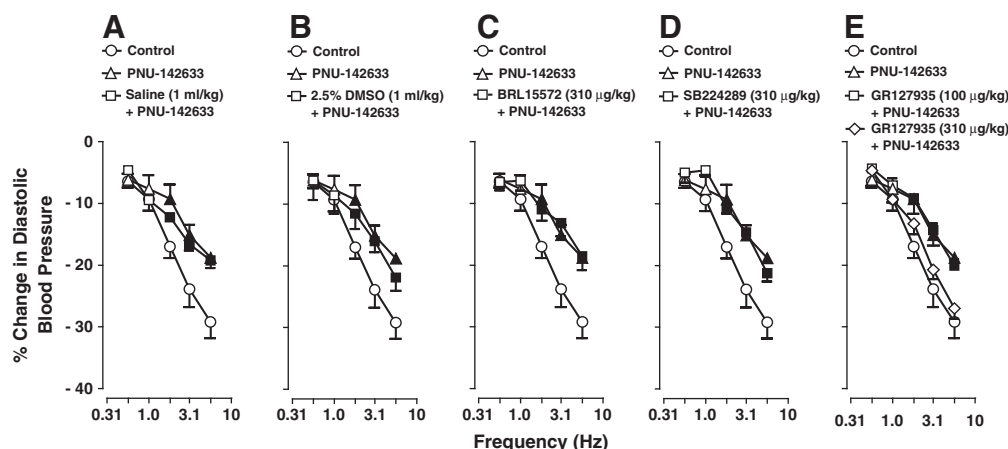


Fig. 8. Effect of i.v. bolus injections of: (A) saline (1 ml/kg; $n=5$); (B) 2.5% DMSO (1 ml/kg; $n=4$); (C) BRL15572 (310 µg/kg; $n=5$); (D) SB224289 (310 µg/kg; $n=5$); or (E) GR127935 (100 and 310 µg/kg; $n=5$ each dose) on the inhibition induced by PNU-142633 (10 µg/kg.min) of the electrically-induced vasodepressor responses. The control group represents that of animals receiving an i.v. methoxamine infusion (20 µg/kg.min), which is shown for comparison. For the sake of clarity, empty symbols depict either control (methoxamine) responses (○) or non-significant responses (△□◇) vs. control. Solid symbols (▲■) represent significantly different responses ($P<0.05$) vs. control.

receptor(s). The same reasoning could be applied to the failure of WAY100635 (5-HT_{1A}; pK_i : 9.6) alone to block sumatriptan-induced inhibition (Fig. 6G). Remarkably, simultaneous blockade of 5-HT_{1A}, 5-HT_{1B} and 5-HT_{1D} receptors with WAY100635 (10 µg/kg) plus GR127935 (100 µg/kg) (Fig. 6H) resembled that with 100 µg/kg GR127935 alone (Fig. 6C). Thus, considering exclusively these results, the involvement of: (i) 5-HT_{1A} receptors seems unlikely; and (ii) 5-HT_{1B}, 5-HT_{1D} and 5-HT_{1F} receptors seems likely (while we have previously shown the role of 5-HT_{1B} receptors; González-Hernández et al., 2010). These suggestions are reinforced with the rank order of selectivity of sumatriptan (i.e. 5-HT_{1D}>5-HT_{1F}>5-HT_{1B}>>5-HT_{1A}; Table 1). The validation of these hypotheses led us to perform additional experiments with selective 5-HT₁ receptor ligands (see below).

4.4. Further evidence excluding the role of sensory-inhibitory 5-HT_{1A} receptors

Further findings excluding the role of 5-HT_{1A} receptors include the lack of effect of NN-DP-5-CT and 8-OH-DPAT (5-HT_{1A} receptor agonists;

Table 1) on the electrically-induced vasodepressor responses (Fig. 3D and E). Higher doses of these agonists could not be used as they produced sustained vasodepressor responses, which represent a problem in our experimental model (Villalón et al., 2008). Interestingly, 31 µg/kg.min indorenate (another 5-HT_{1A} receptor agonist; pK_i : 7.8) did inhibit the vasodepressor sensory outflow (Fig. 3C), but 10 µg/kg WAY100635: (i) failed to block this response (Fig. 7B), a fact that excludes 5-HT_{1A} receptors; and (ii) abolished the sympatho-inhibition to 30 µg/kg.min indorenate in pithed rats, where 5-HT_{1A} receptors are involved (Villalón et al., 1998). Importantly, the fact that indorenate-induced inhibition was: (i) unaffected by 100 µg/kg GR127935 (Fig. 7C) excludes the role of 5-HT_{1B} and 5-HT_{1D} receptors; and (ii) blocked by 310 µg/kg GR127935 (Fig. 7C) suggests the role of 5-HT_{1F} receptors, as GR127935 displays a moderate affinity for these receptors (Table 1). Nevertheless, the affinity of indorenate for 5-HT_{1F} receptors is not known.

The inhibition by 31 µg/kg WAY100635 (Fig. 5D) might involve activation of other mechanisms including prejunctional 5-HT₁ receptors (this study), dopamine D₄ receptors (Marona-Lewicka and Nichols, 2009) and/or α_2 -adrenoceptors (Testa et al., 1999). Evidently, further experiments will be required to evaluate these possibilities.

4.5. Possible involvement of sensory-inhibitory 5-HT_{1B} and 5-HT_{1D} receptors

Since sumatriptan inhibited the vasopressor sensory outflow, 5-HT_{1B} and 5-HT_{1D} receptors could be involved. In fact, selective activation of 5-HT_{1B} receptors inhibits the rat vasodepressor sensory outflow (González-Hernández et al., 2010); our findings reconfirm the role of these receptors as the inhibition to sumatriptan was partly blocked when SB224289 and BRL15572 were given together, but not separately (see above). However, the involvement of 5-HT_{1D} receptors cannot be excluded; this is why PNU-142633 (a 5-HT_{1D} receptor agonist; Table 1) was further evaluated.

Indeed, PNU-142633 inhibited the vasodepressor sensory outflow (Figs. 3F and 4D), but this response was not blocked by 310 µg/kg BRL15572 (Fig. 8C), a dose that completely blocks prejunctional 5-HT_{1D} receptors (Sánchez-López et al., 2004). Furthermore, consistent with its very low affinity for 5-HT_{1B} receptors (Table 1), the inhibition to PNU-142633 remained unaffected after SB224289 (Fig. 8D). However, it could still be argued that PNU-142633 could have simultaneously stimulated 5-HT_{1B} and 5-HT_{1D} receptors; thus when blocking only one of these receptors, the inhibition produced by the unblocked receptor would overshadow the antagonism of the other receptor. But the fact that 100 µg/kg GR127935 failed to block the inhibition by PNU-142633

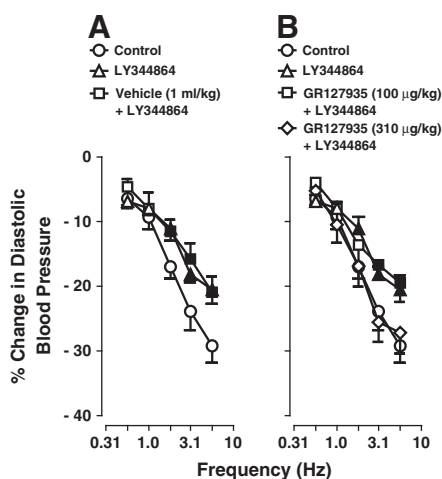


Fig. 9. Effect of i.v. bolus injections of: (A) vehicle (saline, 1 ml/kg); or (B) GR127935 (100 and 310 µg/kg) on the inhibition induced by LY344864 (31 µg/kg.min) ($n=5$ each dose) of the electrically-induced vasodepressor responses. The control group represents that of animals receiving an i.v. continuous infusion of methoxamine (20 µg/kg.min), which is shown for comparison. For the sake of clarity, empty symbols depict either control (methoxamine) responses (○) or non-significant responses (△□◇) vs. control. Solid symbols (▲■) represent significantly different responses ($P<0.05$) vs. control.

(Fig. 8E) excludes the co-involvement of 5-HT_{1B} and 5-HT_{1D} receptors. On this basis, it is tempting to suggest that the inhibition by 10 µg/kg. min PNU-142633 involves other mechanisms although, admittedly, its reported binding affinities are only restricted to 5-HT_{1B} and 5-HT_{1D} receptors (Table 1). Interestingly, the fact that 310 µg/kg GR127935 abolished the inhibition by PNU-142633 (Fig. 8E), as previously observed with indorenate (see above), suggests once again the role of 5-HT_{1F} receptors, as GR127935 displays a moderate affinity for these receptors (Table 1). To validate this suggestion, the effects of a 5-HT_{1F} receptor agonist (LY344864) were analysed.

4.6. Further evidence supporting the role of sensory-inhibitory 5-HT_{1F} receptors

The fact that LY344864, a 5-HT_{1F} receptor agonist (Table 1), inhibited the vasodepressor sensory outflow (Figs. 3G and 4E) implies the role of 5-HT_{1F} receptors. This suggestion is reinforced when considering that: (i) LY344864-induced inhibition was blocked by a supramaximal dose (310 µg/kg) of GR127935 (Fig. 9B), which can antagonise 5-HT_{1F} receptors with reasonable potency (Pauwels et al., 1997; Wainscott et al., 1998); and (ii) a lower dose of GR127935 (100 µg/kg, which blocks 5-HT_{1B/1D} receptors; see above) failed to block this response (Fig. 9B). Nevertheless, we cannot exclude the possible co-involvement of 5-HT_{1B} receptors as LY344864 (31 µg/kg. min) displays a relatively low affinity for these receptors (Table 1). Thus, the role of 5-HT_{1F} receptors will be definitely established with the advent of a selective 5-HT_{1F} receptor antagonist which, to the best of our knowledge, is not yet available.

4.7. Possible locus of the sensory-inhibitory 5-HT_{1B} and 5-HT_{1F} receptors

Lastly, one might speculate upon the possible locus of the 5-HT_{1B} and 5-HT_{1F} receptors mediating inhibition of the vasodepressor CGRPergic sensory outflow. Central or spinal mechanisms can be categorically excluded since pithed rats were used. Hence, the possible locus of the 5-HT_{1B} and 5-HT_{1F} receptors could be at the level of sensory perivascular nerves. Admittedly, we have no direct evidence to support this location, but other findings obtained by molecular biology techniques have shown: (i) the presence of 5-HT_{1B} and 5-HT_{1F} receptor mRNA and protein in rat dorsal root ganglia neurons (Chen et al., 1998; Nicholson et al., 2003); and (ii) the absence of 5-HT_{1A} and 5-HT_{1D} receptors in small diameter neurons in rat dorsal root ganglia (Nicholson et al., 2003).

4.8. Final considerations on our experimental conditions and the use of 5-HT

As previously discussed (González-Hernández et al., 2010), 5-HT (endogenous ligand) is not appropriate to identify 5-HT₁ receptors in our experimental model as it stimulates 5-HT₇ receptors producing hypotension (Villalón and Centurión, 2007). This effect of 5-HT is an impediment to determine the vasodepressor sensory-inhibition by 5-HT, as diastolic blood pressure must be kept around 135 mm Hg (Villalón et al., 2008).

5. Conclusion

In addition to excluding the role of 5-HT_{1A} and 5-HT_{1D} receptors, our results in pithed rats: (i) further suggest that activation of prejunctional 5-HT_{1F} receptors inhibits the vasodepressor sensory CGRPergic outflow; and (ii) reconfirm the involvement of 5-HT_{1B} receptors, as previously reported (González-Hernández et al., 2010).

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